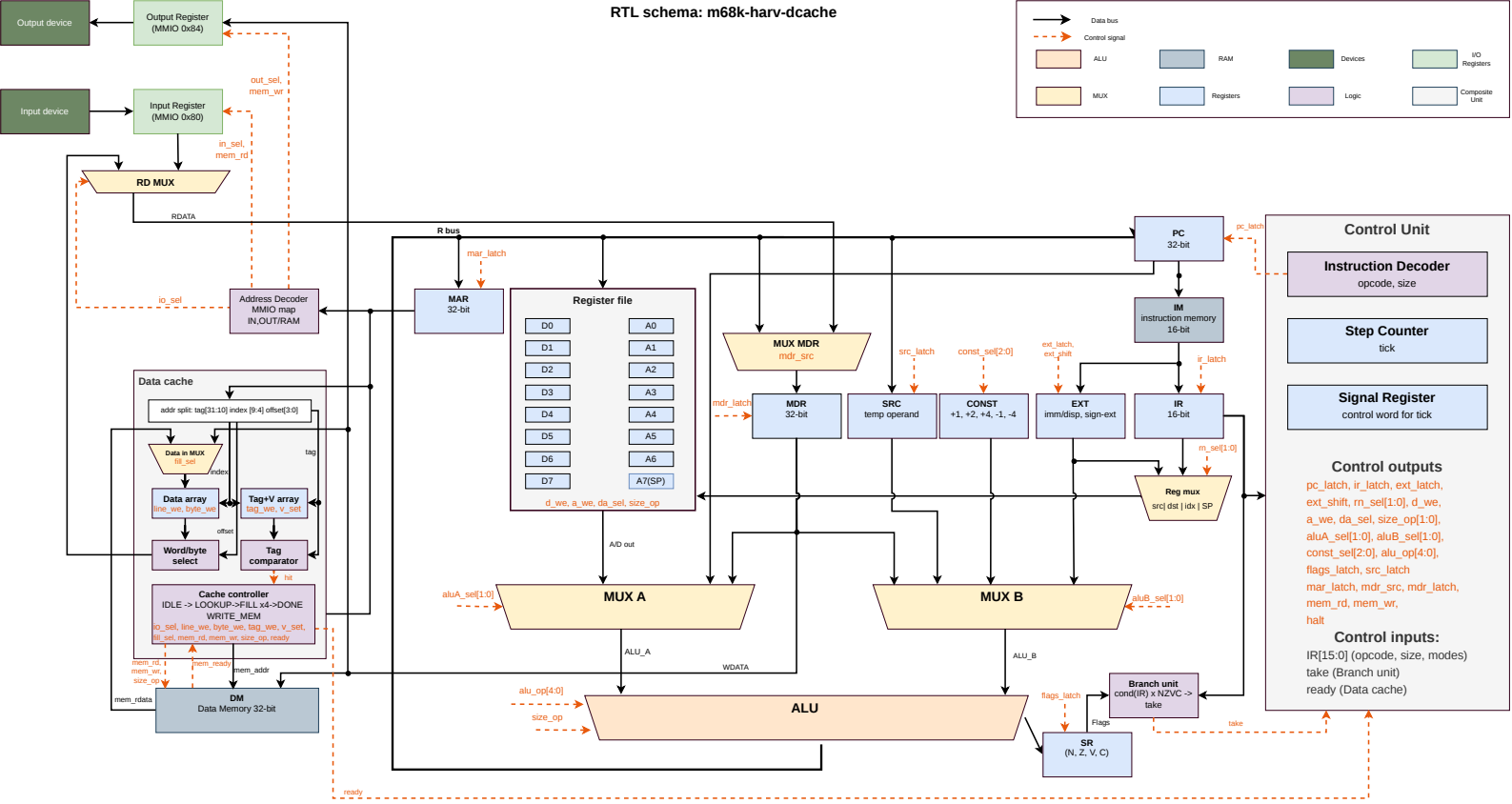

RTL schema: m68k-harv-dcache
Data bus
Control signal
ALU
RAM
Devices
I/O Registers
MUX
Registers
Logic
Composite Unit
Output device
Output Register (MMIO 0x84)
out_sel, mem_wr
Input device
Input Register (MMIO 0x80)
in_sel, mem_rd
RD MUX
RDATA
R bus
mar_latch
io_sel
Address Decoder MMIO map IN,OUT/RAM
MAR 32-bit
Register file
D0
D1
D2
D3
D4
D5
D6
D7
A0
A1
A2
A3
A4
A5
A6
A7(SP)
d_we, a_we, da_sel, size_op
PC 32-bit
pc_latch
IM instruction memory 16-bit
ir_latch
MUX MDR mdr_src
mdr_latch
src_latch
const_sel[2:0]
ext_latch, ext_shift
MDR 32-bit
SRC temp operand
CONST +1, +2, +4, -1, -4
EXT imm/disp, sign-ext
IR 16-bit
rn_sel[1:0]
Reg mux src| dst | idx | SP
Data cache
addr split: tag[31:10] index [9:4] offset[3:0]
Data in MUX fill_sel
tag
index
offset
Data array line_we, byte_we
Tag+V array tag_we, v_set
Word/byte select
Tag comparator
hit
Cache controller
IDLE -> LOOKUP->FILL x4->DONE
WRITE_MEM
io_sel, line_we, byte_we, tag_we, v_set, fill_sel, mem_rd, mem_wr, size_op, ready
A/D out
aluA_sel[1:0]
MUX A
MUX B
aluB_sel[1:0]
ALU_A
ALU_B
WDATA
mem_rd, mem_wr, size_op
mem_ready
mem_addr
mem_rdata
DM Data Memory 32-bit
alu_op[4:0]
size_op
ALU
flags_latch
Flags
Branch unit cond(IR) x NZVC -> take
take
SR (N, Z, V, C)
ready
Control Unit
Instruction Decoder
opcode, size
Step Counter
tick
Signal Register
control word for tick
Control outputs
pc_latch, ir_latch, ext_latch, ext_shift, rn_sel[1:0], d_we, a_we, da_sel, size_op[1:0], aluA_sel[1:0], aluB_sel[1:0], const_sel[2:0], alu_op[4:0], flags_latch, src_latch mar_latch, mdr_src, mdr_latch, mem_rd, mem_wr, halt
Control inputs:
IR[15:0] (opcode, size, modes)
take (Branch unit)
ready (Data cache)